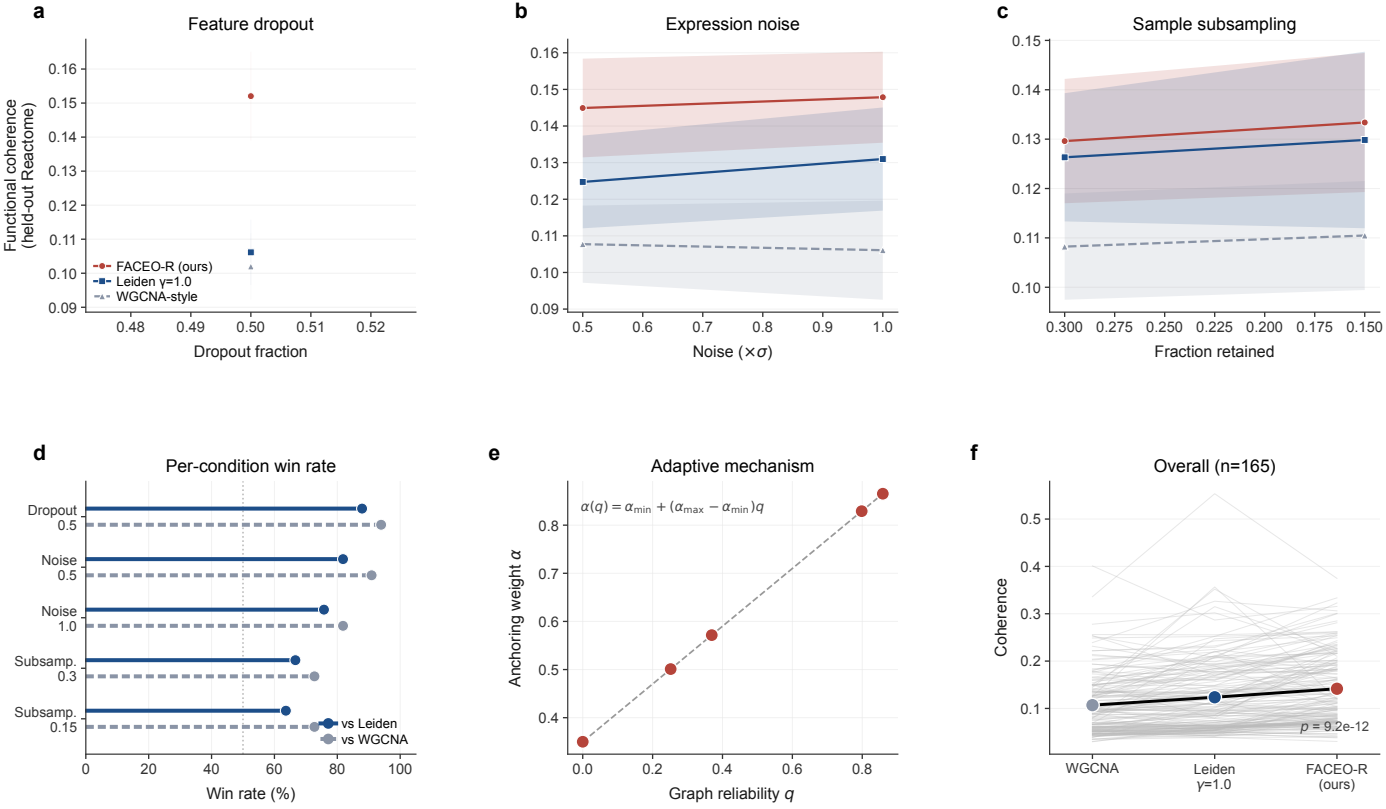

a
Feature dropout
Functional coherence (held-out Reactome)
0.16
0.15
0.14
0.13
0.12
0.11
0.10
0.09
FACEO-R (ours)
Leiden γ=1.0
WGCNA-style
0.48
0.49
0.50
0.51
0.52
Dropout fraction
b
Expression noise
0.16
0.15
0.14
0.13
0.12
0.11
0.10
0.09
0.5
0.6
0.7
0.8
0.9
1.0
Noise (×σ)
c
Sample subsampling
0.15
0.14
0.13
0.12
0.11
0.10
0.300
0.275
0.250
0.225
0.200
0.175
0.150
Fraction retained
d
Per-condition win rate
Dropout 0.5
Noise 0.5
Noise 1.0
Subsamp. 0.3
Subsamp. 0.15
vs Leiden
vs WGCNA
0
20
40
60
80
100
Win rate (%)
e
Adaptive mechanism
$\alpha(q) = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min})q$
Anchoring weight α
0.8
0.7
0.6
0.5
0.4
0.0
0.2
0.4
0.6
0.8
Graph reliability q
f
Overall (n=165)
Coherence
0.5
0.4
0.3
0.2
0.1
p = 9.2e-12
WGCNA
Leiden γ=1.0
FACEO-R (ours)